

Mini Report

1. I mainly used methods such as extracting functions, renaming variables, decomposing conditional statements and code reuse to remove duplicate freight logic.
2. The original code has problems such as unclear variable naming, logical repetition, and the long `generate_report()` method (which confuses multiple responsibilities), so these refactorings make the code easier to read and maintain.
3. I ran unit tests before and after the reconstruction, and then added additional tests to the branches that were not covered before. All 13 tests have passed, and the code coverage of `main.py` has reached 100%.
4. The advantage of this is that the code is easier to read, the structure is clearer, the code is less repetitive, and future modifications are safer. A possible disadvantage is that the code now has more auxiliary methods, which may appear a little fragmented in smaller files.
5. Yes, it's worth doing this. The behaviour of the program remains unchanged, but the code is more concise, easier to expand, and has better test protection.